

Global Energy Storage

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Battery Weekly 1 June

America

- [LG Energy Solution Signs \\$1.6 Billion ESS Supply Deal With DTE Energy.- thelec.net](#) LG Energy Solution signed a \$1.6 billion ESS battery supply agreement with U.S. utility DTE Energy, covering 6 GWh of batteries over about two years. The batteries will support eight grid projects in Michigan, including an AI data center development, and will be produced at LGES's Holland, Michigan facility. The deal highlights growing demand for ESS in grid stabilization and AI infrastructure, while reinforcing LGES's strategy to expand its North American production footprint, targeting over 60 GWh global ESS capacity by year-end.
- [LG Energy Solution sells Honda JV battery plant assets.- pulse.mk](#) LG Energy Solution has completed the sale of building assets from its U.S. battery joint venture with Honda for KRW 3.74 trillion (USD 2.48 billion), securing additional liquidity amid a slowdown in EV demand. The assets, located at the Ohio battery plant, were sold to Honda's U.S. unit but will continue to be used through lease arrangements, ensuring operational continuity. The move reduces LGES's short-term capital burden while improving cash flow, with the proceeds expected to support the joint venture's efficiency. The plant is set to begin full-scale operations this year, supplying batteries for Honda and Acura EVs in North America, with potential expansion into hybrid vehicles and ESS.

Asia

- [SK Innovation to Sell Loss-Making Chinese Separator Business to Semcorp for USD69 Million.- yicaiglobal](#) SK Innovation will sell its loss-making China battery separator business (SK Hi-Tech Materials) to Semcorp for about CNY 400 million (USD ~59 million) as part of its restructuring. The Changzhou plant has ~940 million sqm capacity but has struggled with low utilization and losses due to limited customers. The acquisition allows Semcorp to quickly expand capacity and improve efficiency using its scale, customer network, and cost advantages, while SK shifts its separator operations toward Poland and other core regions.
- [China's Tinci Gains on Plan to Build Lithium Battery Material Plant for Up to USD310 Million.- yicaiglobal](#) Tinci Materials Technology plans to invest up to CNY 2.1 billion (~USD 310 million) to build a new LFP cathode material plant in Zhejiang, China, with annual capacity of 160,000 tonnes. The project focuses on high-compaction LFP materials, which offer higher energy density, better fast-charging, and longer cycle life, targeting both high-end EV batteries and large-scale energy storage systems. The expansion is part of Tinci's strategy to move beyond electrolytes into integrated battery materials, strengthening its supply chain position for customers such as CATL, BYD, and Gotion.
- [China opens first truck battery swap station using Bosch BaaS.- electrive.com](#) A new electric truck battery-swapping station has been launched in Chizhou, China, using a Battery-as-a-Service (BaaS) solution from the Bosch-Mitsubishi joint venture Bosch MC Battery Service Innovations. The station, developed with Shanghai Lingzhou, can serve over 100 trucks per day and combines swapping with AI-based battery monitoring and cloud analytics to track battery health and optimize performance. The project demonstrates growing adoption of BaaS models in commercial fleets, helping reduce downtime, manage battery degradation, and improve total cost of ownership as China accelerates electrification of heavy-duty transport.
- [CATL launches battery-swapping network for light electric trucks in southern China.- electrive.com](#) CATL has partnered with logistics provider DST to launch its first standardized battery-swapping network for light electric trucks in China's Greater Bay Area. The project currently operates 31 swap stations and plans to expand to 140 sites by the end of 2026, supporting around 5,000 electric trucks. Using CATL's Choco-SEB battery-swapping system, the network enables fully automated battery swaps in about 2 minutes, reducing refueling

time and cutting operating costs by roughly 50% compared to diesel trucks. The rollout is part of CATL's broader strategy to build a nationwide network of 3,000 swap stations and accelerate electrification of commercial transport.

- [POSCO Future M to Build LFP Cathode Plant in Pohang.- thelec.net](#) POSCO Future M, through its joint venture CNP New Material Technology, is building a new LFP cathode materials plant in Pohang, South Korea, with production set to begin in 2027 and capacity expanding to 50,000 tonnes annually. The move reflects rising demand for cost-efficient LFP batteries, particularly for energy storage systems and AI-driven power demand. In parallel, POSCO Future M is also converting part of its existing ternary cathode lines to LFP production, starting pilot runs this year. The investment supports the company's strategy to diversify its cathode portfolio beyond high-nickel materials and strengthen competitiveness in global battery markets.
- [Hyosung Heavy Industries Wins 11 Billion Won ESS Order in Japan.- thelec.net](#) Hyosung Heavy Industries has secured an ESS EPC contract worth about KRW 11 billion in Japan to build 10 MW / 40 MWh grid-connected energy storage systems across five regions. The company will also provide up to 20 years of maintenance services, strengthening its long-term presence in Japan's ESS market. Including this deal, Hyosung has secured around KRW 64 billion in orders in its first year in Japan, following a larger project in Hokkaido earlier this year.
- [China's SINVO Starts Full Operations at Korea Plant for Tesla Battery Equipment.- thelec.net](#) SINVO (Shenzhen Sinvo Automation) has established a battery equipment manufacturing plant in Gumi, South Korea, to supply LG Energy Solution, with equipment ultimately destined for LG's LFP battery production lines at its Lansing, Michigan plant for Tesla's Megapack ESS. The factory focuses on stacking process equipment, a critical and technically demanding step in battery manufacturing that affects performance and energy density. Most of the current workforce consists of Chinese staff, with plans to scale operations as demand increases. The move reflects a broader trend of Chinese battery equipment makers expanding into Korea, supporting global battery supply chains tied to large-scale ESS projects and Tesla-related production.
- [JR Energy Solution Partners With Factorial on Solid-State Drone Batteries.- thelec.net](#) JR Energy Solution (JRES) has partnered with U.S.-based Factorial Energy to commercialize solid-state and lithium-metal batteries for drones, aiming to improve energy density, range, and performance. JRES will handle battery manufacturing across the Asia-Pacific region, leveraging its full in-house capabilities from electrodes to battery packs, while Factorial contributes next-generation battery technology. The collaboration is part of a global three-region strategy (U.S., Europe, Asia) and reflects growing demand for advanced batteries in drones, robotics, and industrial mobility.
- [From Tesla to BYD, Chinese-made EVs capture one-third of Korea's market.- korea joongang daily](#) Chinese-made EVs are rapidly gaining market share in South Korea, rising to 33.9% of new EV registrations in 2025 from just 1.1% in 2021, while domestic brands fell to 57.2%. Much of the growth is driven by Tesla's China-built vehicles, but Chinese brands like BYD are also expanding quickly with lower-priced models. The surge highlights growing competition driven by cost advantages, as Korea maintains relatively low tariffs, making it harder for domestic automakers and battery suppliers to compete.
- [Zeron Raises USD200 Million as China's Electric Heavy Truck Race Heats Up.- yicaiglobal](#) Chinese electric heavy-duty truck startup Zeron has raised USD 200 million in a new funding round, bringing its total fundraising to USD 400 million in just two months, as competition intensifies in the electric commercial vehicle market. The round attracted major industrial investors such as Zijin Mining, Shandong Energy Group, and Sanhua, along with global investors including Temasek. Founded in 2022, Zeron has rapidly scaled sales of its electric truck models and achieved positive operating cash flow, with the new funding aimed at accelerating large-scale deployment, autonomous driving development, and global expansion in the heavy-duty EV sector.
- [NR Electric Delivers World's Largest Grid-Forming Energy Storage Cluster in Saudi Arabia \(2.5 GW\).- cn.solarbe](#) NR Electric has delivered and commissioned a 2.5 GW grid-forming energy storage project for Saudi Electricity Company (SEC), now the largest grid-scale grid-forming ESS cluster in the world. The project spans five major regions—Riyadh, Rabigh, Dawadmi, Al-Jouf, and Qaisumah—and includes more than 2,000 power conversion systems (PCS). It supports advanced grid functions such as frequency regulation, voltage control, black start capability, and weak-grid stabilization, playing a key role in strengthening Saudi Arabia's power system. Despite a tight delivery timeline of less than four months, NR Electric completed system design in one week and dispatched the first integrated storage units within 53 days, highlighting rapid execution and large-scale deployment capability in utility-scale ESS projects.

Europe

- [Orbia secures £1.4 million UK grant for graphite recycling pilot.- electrive.com](#) Orbia Fluor & Energy Materials has secured £1.4 million in UK government funding to develop a graphite recycling pilot project in Runcorn, aimed at strengthening the domestic EV battery supply chain. The project will demonstrate graphite recovery technology at a small scale before potential industrial expansion, helping reduce reliance on imports—particularly from China, which dominates global graphite supply. Backed by the UK's DRIVE35 program, the initiative supports building a local, circular battery materials ecosystem and advancing sustainable battery production.

- [DHL Supply Chain opens battery logistics hub in the Netherlands.- electrive.com](#) DHL Supply Chain is building a dedicated high-voltage battery logistics hub in Holtum, Netherlands, with operations expected to begin in early 2027. The 17,000 m² facility will focus exclusively on batteries for electric vehicles and energy storage systems (BESS), forming part of a broader European electromobility logistics campus. The center will offer end-to-end battery lifecycle services, including storage, testing, charging, maintenance, reverse logistics, and recycling preparation, strengthening supply chain support for the growing EV and ESS markets.
- [Basquevolt launches lithium-metal battery cell.- electrive.com](#) Basquevolt has launched its new BQV400L lithium-metal battery cell, marking its first commercially available high-energy-density product. The cell combines an NMC cathode, lithium-metal anode, and hybrid polymer electrolyte, delivering around 402 Wh/kg, positioning it among the highest-performance cells produced in Europe. Designed as a “drop-in” solution compatible with existing gigafactory production lines, the battery aims to accelerate industrial adoption without major additional investment. The technology represents a step toward next-generation solid-state batteries and targets applications in EVs, aerospace, and industrial sectors, with partners such as Renault supporting further development.

EXHIBIT 1: **Key commodities price performance**

	Price 29-May	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	25,629					1.3%	-2%	2%	206%	
LiOH (spot)	24,521					0.1%	-3%	1%	180%	
LiCO (contract)	20,250					0.0%	-7%	8%	108%	
LiOH (contract)	20,000					0.0%	1%	10%	135%	
Cobalt (spot)	55,858					0.0%	0%	0%	68%	
Nickel (spot)	19,101					0.8%	1%	-2%	24%	
Component prices										
Cathode NMC811						0.0%	-1%	4%	53%	
Cathode LFP						1.2%	0%	2%	92%	
Precursor NMC811						0.0%	0%	2%	37%	
Precursor LFP						0.4%	0%	8%	30%	
Artificial Graphite						0.0%	0%	2%	4%	
Natural Graphite						0.0%	0%	0%	-6%	
Electrolyte NMC						0.0%	0%	-3%	23%	
Electrolyte LFP						0.0%	0%	-3%	43%	
Separator						0.0%	2%	5%	10%	
Spot battery cell cost (US\$/kWh)										
LFP	62					0.3%	0%	0%	19%	
NMC532	91					0.0%	0%	3%	31%	
NMC622	89					0.0%	-1%	3%	28%	
NMC811	88					0.0%	0%	2%	25%	
Spot battery pack cost (US\$/kWh)										
LFP	76					0.3%	0%	-1%	3%	
NMC532	122					0.0%	0%	2%	21%	
NMC622	122					0.0%	0%	2%	19%	
NMC811	121					0.0%	0%	1%	17%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price				EV					
	29-May	Ccy	Mkt cap	P/E	/EBITDA	%1D	%1W	%1M	%1Y	1Y px
			\$bn	26E	26E					chart
Cell makers										
LGES	457,500	KRW	71.3	720.5x	18.8x	2.9%	14%	-2%	59%	
Samsung SDI	683,000	KRW	36.6	178.0x	23.8x	1.5%	6%	1%	292%	
CATL	424.4	CNY	296.8	20.4x	14.4x	2.1%	3%	-1%	68%	
CALB	29.4	HKD	7.2	17.5x	5.1x	-4.1%	-1%	-9%	79%	
EVE Energy	64.2	CNY	20.6	19.0x	12.2x	-2.5%	0%	-9%	41%	
Gotion	33.3	CNY	8.9	27.8x	8.0x	-3.3%	-6%	-15%	34%	
Sunwoda	22.9	CNY	6.3	16.4x	6.3x	-3.9%	-6%	-13%	22%	
Farasis	13.1	CNY	2.4	n.a.	n.a.	-5.5%	-11%	-4%	0%	
Lithium Miners/Refiners										
Albemarle	176.3	USD	20.8	14.9x	7.5x	-0.6%	3%	-6%	207%	
SQM	84.1	USD	22.6	13.6x	6.9x	0.1%	5%	-6%	167%	
Pilbara Minerals	6.4	AUD	14.7	38.2x	19.3x	1.2%	1%	4%	385%	
Ganfeng Lithium	64.1	HKD	21.1	16.2x	13.8x	-0.6%	-6%	-18%	229%	
Tianqi Lithium	63.1	CNY	15.3	19.6x	6.5x	-2.0%	-5%	-13%	115%	
Cathode										
LG Chem	363,500	KRW	17.1	177.1x	3.4x	6.1%	4%	-7%	82%	
EcoPro BM	214,500	KRW	14.0	431.8x	78.7x	0.2%	-2%	2%	138%	
L&F	164,600	KRW	4.4	1622.7x	25.3x	-3.2%	2%	-19%	189%	
Umicore	25.8	EUR	7.4	15.8x	6.6x	0.7%	-1%	50%	169%	
Ningbo Ronbay	31.1	CNY	3.3	49.8x	14.1x	-3.9%	-11%	-12%	68%	
Beijing Easpring	55.8	CNY	4.5	28.5x	18.2x	-3.4%	-7%	-10%	45%	
GEM	7.8	CNY	5.9	18.5x	5.9x	-2.5%	-6%	-13%	27%	
Huayou Cobalt	55.2	CNY	15.5	11.8x	5.6x	-1.3%	-6%	-15%	63%	
Dynanonic	66.4	CNY	2.7	38.7x	12.9x	-2.4%	-10%	25%	135%	
Anode										
Posco	241,500	KRW	14.3	510.1x	59.9x	2.1%	-1%	-7%	120%	
Putailai	30.1	CNY	9.5	19.4x	12.3x	-1.8%	-7%	-15%	86%	
Shanshan	14.8	CNY	4.9	n.a.	n.a.	-3.0%	-1%	-6%	94%	
Separator										
Asahi Kasei	1,790	JPY	15.3	16.6x	6.1x	2.0%	4%	16%	76%	
SK IE	19,250	KRW	1.0	n.a.	-45.8x	-1.2%	-7%	-28%	-9%	
Toray Industries	1,186	JPY	11.2	22.1x	6.8x	4.7%	9%	4%	18%	
W-Scope	291	JPY	0.1	n.a.	n.a.	-3.6%	-15%	30%	3%	
Yunnan Energy	70.3	CNY	10.2	40.6x	14.5x	-2.1%	-4%	-10%	149%	
Shenzhen Senior	15.9	CNY	3.2	35.1x	10.8x	-5.9%	-13%	1%	59%	
Sinoma Science	71.6	CNY	17.7	44.3x	17.9x	-6.4%	0%	34%	319%	
Electrolyte										
Enchem	34,750	KRW	0.5	n.a.	n.a.	-2.0%	-11%	-27%	-46%	
Tinci Minerals	52.6	CNY	15.8	15.4x	11.1x	1.3%	-10%	-3%	206%	
Capchem	74.2	CNY	8.3	27.3x	17.2x	-7.2%	2%	12%	141%	
Jiangsu Guotai	8.9	CNY	2.1	11.4x	n.a.	0.5%	-1%	-7%	20%	
Shanshan	14.8	CNY	4.9	n.a.	n.a.	-3.0%	-1%	-6%	94%	
Equipment Maker										
Hirano	2,031	JPY	0.2	19.2x	n.a.	9.5%	29%	14%	38%	
CKD	6,700	JPY	2.9	36.0x	18.6x	-1.5%	-1%	14%	186%	
Wuxi Lead	51.0	CNY	12.5	33.6x	26.9x	-0.9%	-1%	-8%	154%	
Shenzhen Yinghe	27.2	CNY	2.6	28.8x	14.1x	-3.9%	-8%	0%	55%	
Zhejiang HangKe	39.4	CNY	3.5	41.4x	39.0x	-0.2%	12%	26%	128%	
Guangdong Lyric	58.0	CNY	1.4	63.7x	23.4x	-5.0%	0%	9%	103%	
Shenzhen United	27.7	CNY	1.4	24.8x	17.0x	-9.7%	-11%	3%	70%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

[Global Energy Storage: Resistance is Futile](#)

Initiation Report

[Global Energy Storage: Initiating Coverage. Lithium Positive \(Tianqi\) and Negative Cathodes \(EcoPro BM, Posco Future M\)](#)

[Sungrow: Solar Eclipse - Initiating at Market-Perform \(PT RMB65\)](#)

Recent Research

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[EV TRACKER - 2025: Solid year of 20% growth, but expectations are for slower 2026](#)

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[Future of Tech: Can China achieve AI supremacy?](#)

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[The Long View: Batteries to only get better](#)

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[Tianqi Lithium: 1Q26 Best Idea - Energy storage powers lithium upside. Outperform](#)

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[Global Energy Storage: 2026 Outlook. Power storage set to continue to drive the battery rally](#)

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[Global Energy Storage: Should investors chase the rally in battery stocks?](#)

[Global Energy Storage: Why is ESS demand booming?](#)

[Global Energy Storage: Increasing CATL price target to CNY420 on blockbuster year for ESS](#)

[China EV & Battery Tour Takeaways, 2025 Edition: Steady Evolution](#)

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[Global Energy Storage: More downside risks ahead for Korean cathode makers](#)

[CATL: Does CATL deserve a premium or discount for HK IPO?](#)

[CATL: Initial perspectives on HK IPO Plans](#)

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[Global Energy Storage: Gigafactory expansion still strong but investment slowing](#)

[Global Energy Storage: Bigger and better batteries enables the rise of the 700km electric vehicle](#)

[Global Energy Storage: 2025 Lithium Outlook. Rising from rock bottom](#)

[Global Energy Storage: Outlook 2025. New markets to drive battery growth](#)

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	29 May 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	424.00	620.00	22.7%	CNY	16.14	21.56	26.02	26.3	19.7	16.3
3750.HK (CATL)	M	HKD	737.00	600.00	93.4%	CNY	16.14	21.56	26.02	39.4	29.5	24.5
051910.KS (LG Chem)	M	KRW	342,500	298,000	38.1%	KRW (13,258.70)	2,042.47	31,118		(25.8)	167.7	11.0
373220.KS (LGES)	M	KRW	444,500	347,000	11.3%	KRW (5,308.10)	1,811.04	9,452.97		(83.7)	245.4	47.0
006400.KS (SDI)	M	KRW	673,000	520,000	247.0%	KRW (9,933.80)	2,099.51	18,376		(67.7)	320.6	36.6
003670.KS (Posco Future M)	U	KRW	236,500	190,000	75.4%	KRW (2,740.96)	376.72	898.20		(86.3)	627.8	263.3
247540.KS (EcoPro BM)	U	KRW	214,000	140,000	92.5%	KRW	403.00	854.00	1,963.00	531.0	250.6	109.0
300274.CH (Sungrow)	O	RMB	177.99	185.00	144.1%	RMB	6.55	8.22	9.33	27.2	21.6	19.1
002466.CH (Tianqi Lithium)	O	CNY	62.88	73.00	67.1%	CNY	0.28	3.29	6.18	222.8	19.1	10.2
9696.HK (Tianqi Lithium)	O	HKD	49.06	61.00	50.3%	CNY	0.28	3.29	6.18	150.1	12.9	6.9
ASIA			2,010.20									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

003670.KS base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

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